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Wu et al.

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(54) **SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF**

(71) Applicant: **CHANGXIN MEMORY TECHNOLOGIES, INC.**, Hefei (CN)

(72) Inventors: **Gongyi Wu**, Hefei (CN); **Xinran Liu**, Hefei (CN); **Yachao Xu**, Hefei (CN); **Longyang Chen**, Hefei (CN)

(73) Assignee: **CHANGXIN MEMORY TECHNOLOGIES, INC.**, Hefei (CN)

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(30) **Foreign Application Priority Data**

Mar. 17, 2022 (CN) 202210266469.4

(51) **Int. Cl.**
H10B 12/00 (2023.01)

(52) **U.S. Cl.**
CPC ***H10B 12/315*** (2023.02); ***H10B 12/482*** (2023.02)

(58) **Field of Classification Search**
CPC H10B 12/482; H10B 12/315
See application file for complete search history.

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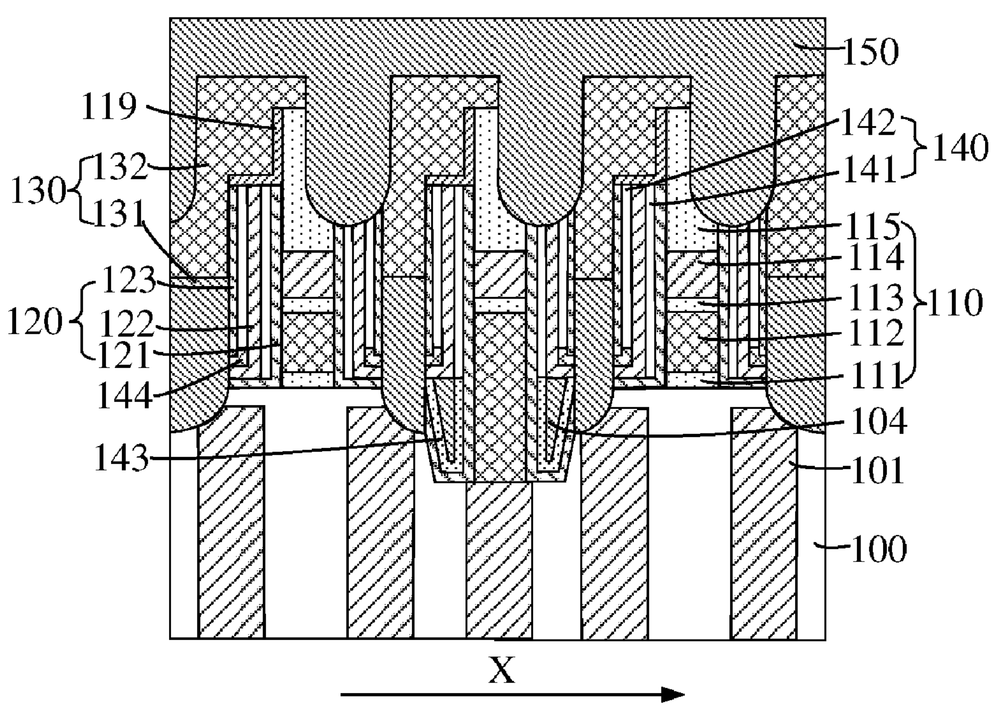
Primary Examiner — Steven M Page
(74) *Attorney, Agent, or Firm* — Kilpatrick Townsend & Stockton LLP

(57) **ABSTRACT**

Embodiments of the present disclosure relate to the field of semiconductor structures, and provide a semiconductor structure and a manufacturing method thereof. The semiconductor structure includes: a base; a plurality of first conductive structures, located on a surface of the base and distributed at intervals along a first direction; a plurality of second conductive structures, located on the surface of the base, and the plurality of second conductive structures and the plurality of first conductive structures being arranged alternately; and a plurality of support structures, located on the surface of the base and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures.

18 Claims, 17 Drawing Sheets



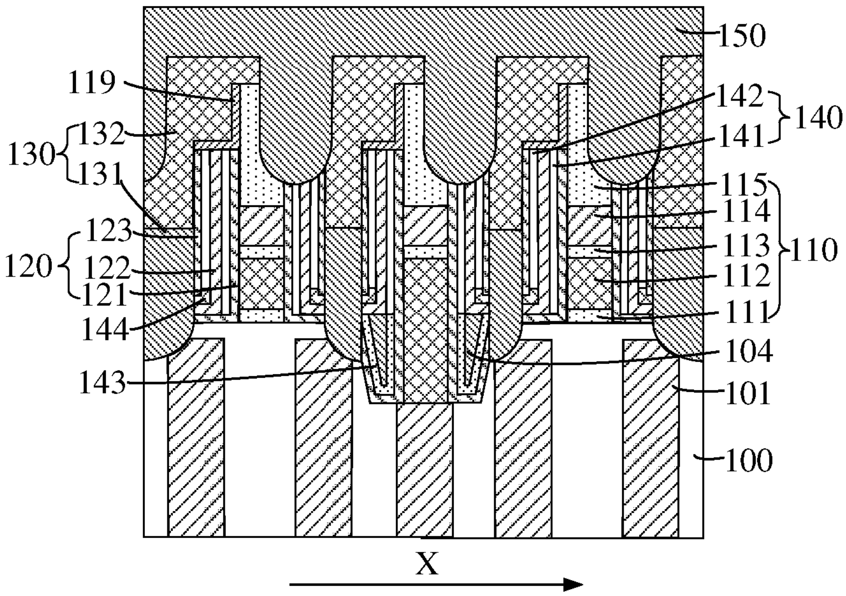

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FIG. 1

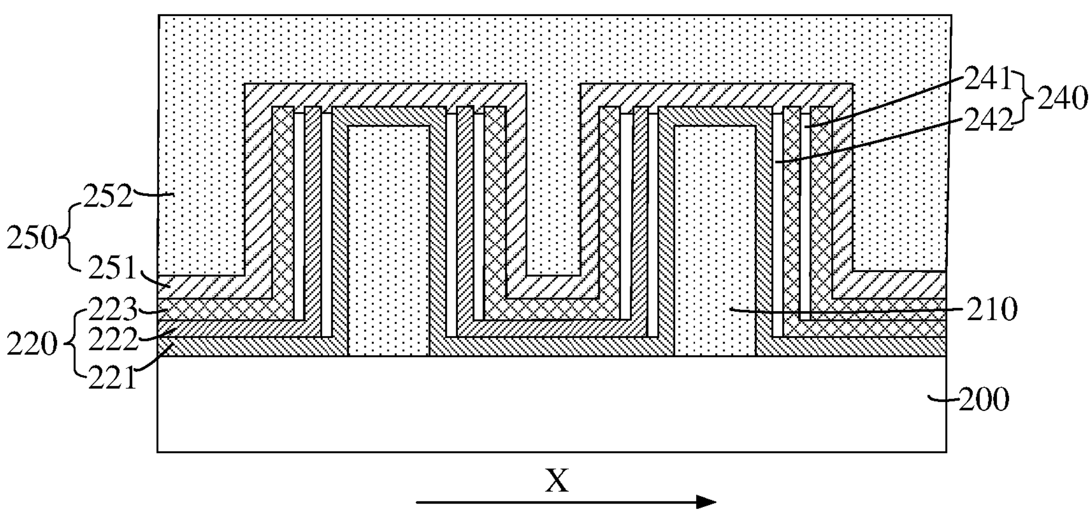
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FIG. 2

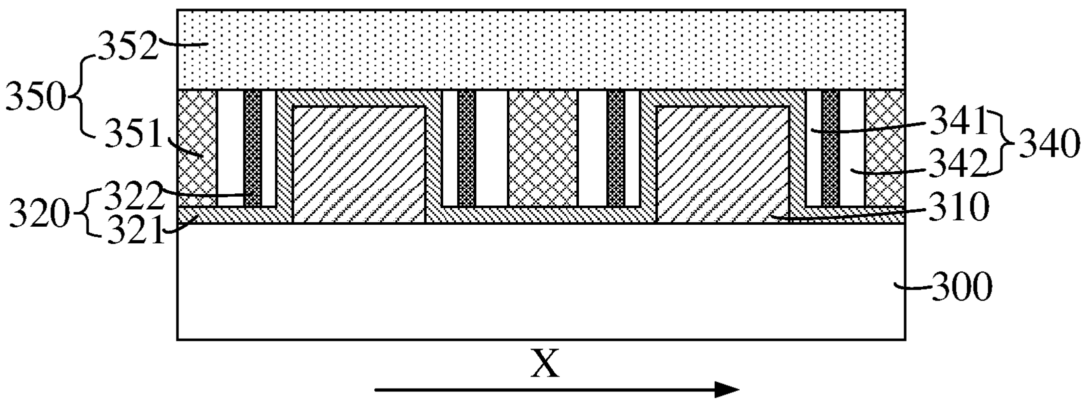

FIG. 3

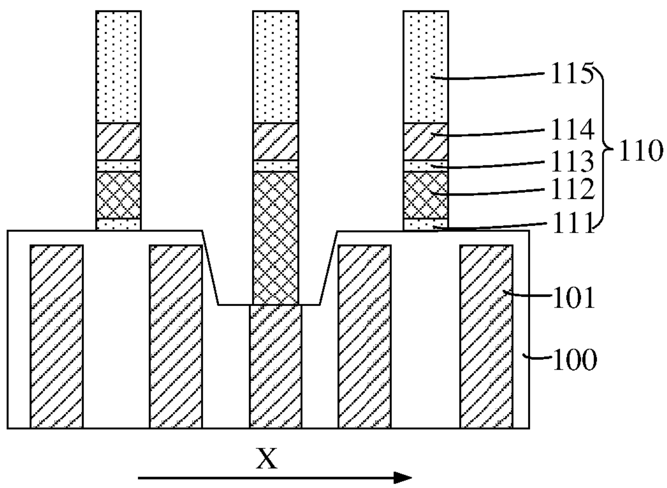

FIG. 4

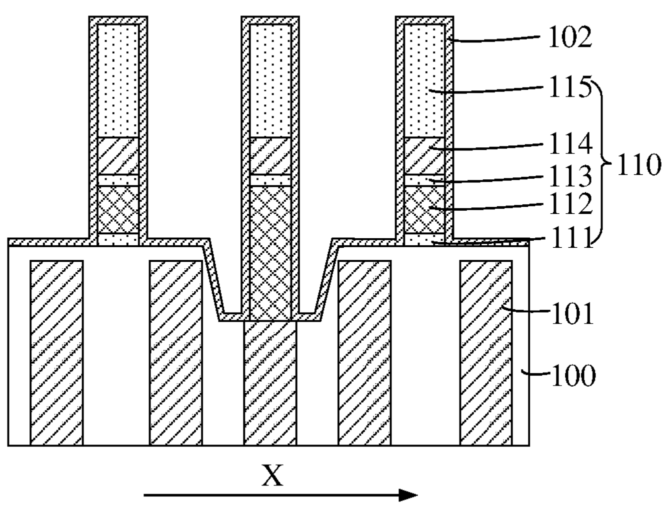

FIG. 5

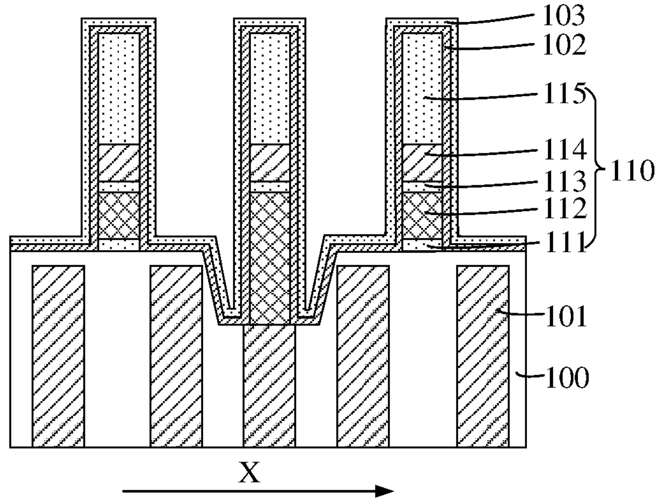
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FIG. 6

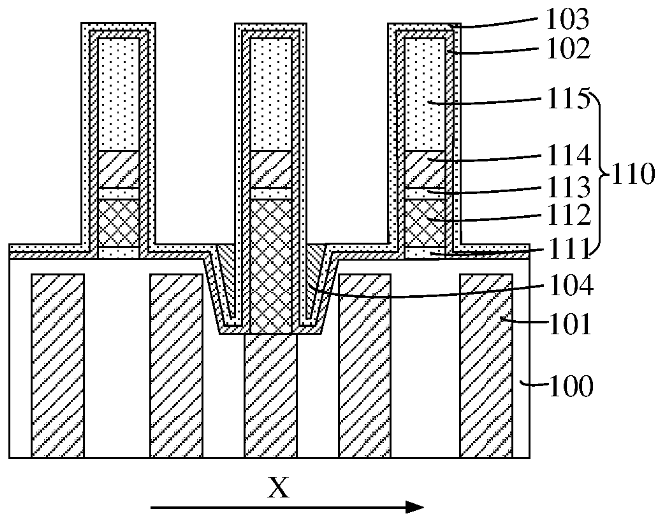
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FIG. 7

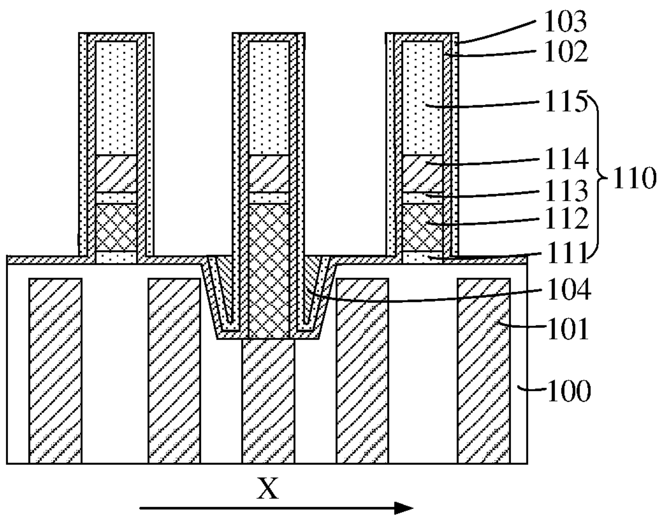
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FIG. 8

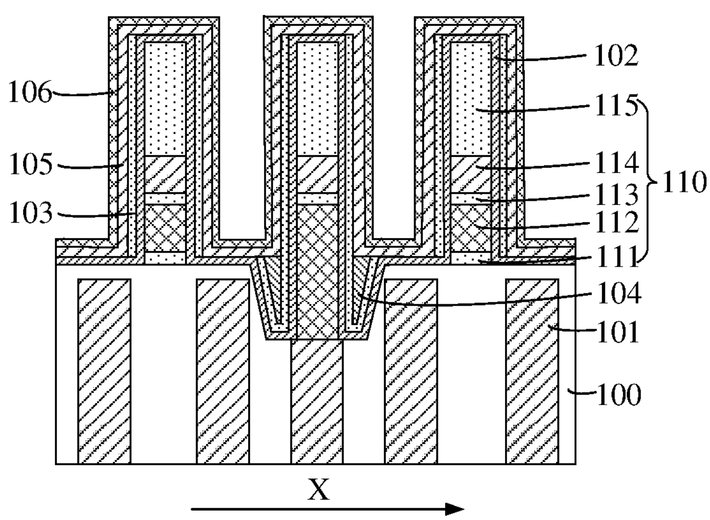
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FIG. 9

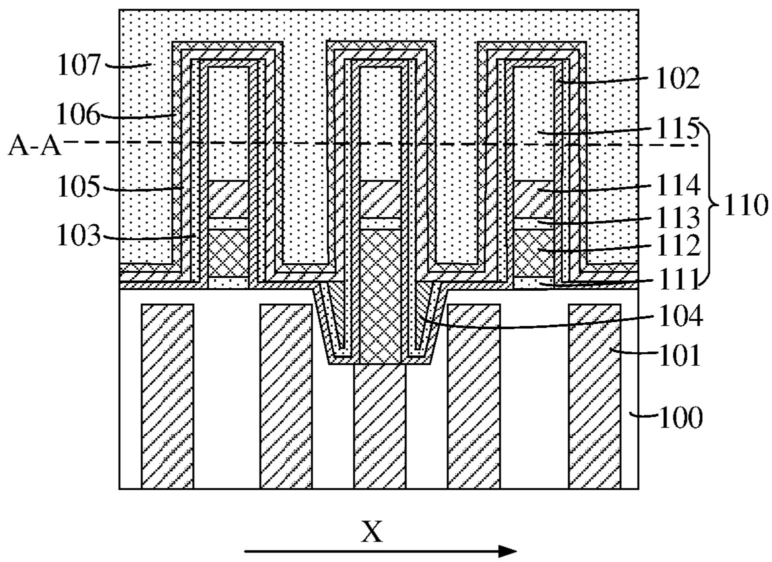
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FIG. 10

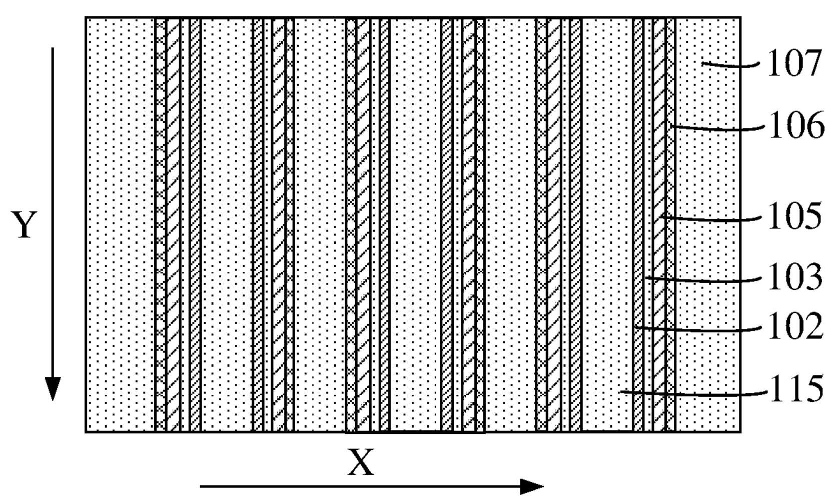
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FIG. 11

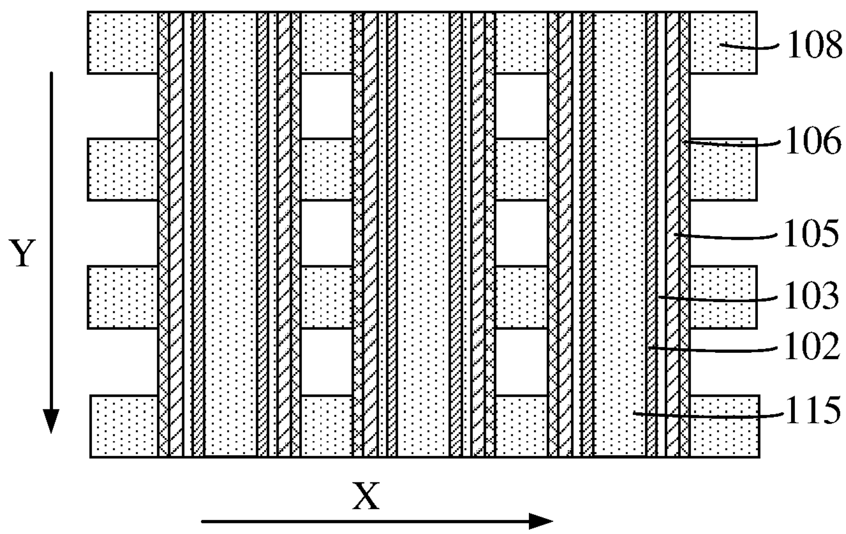
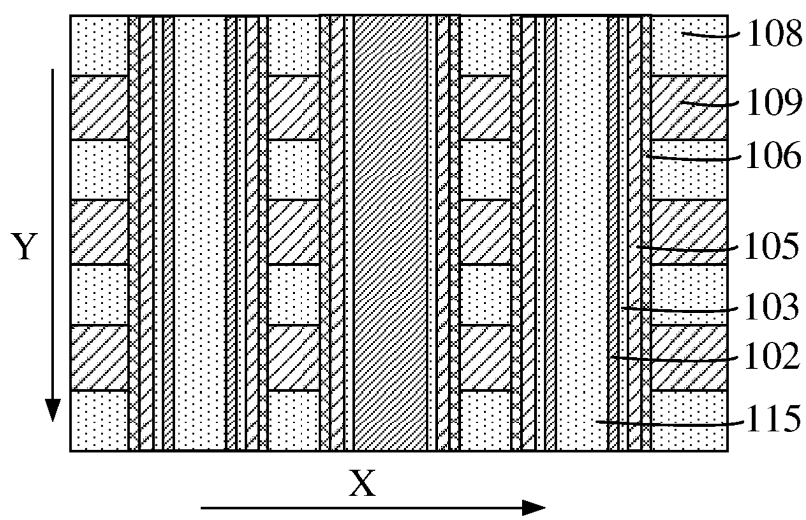


FIG. 12



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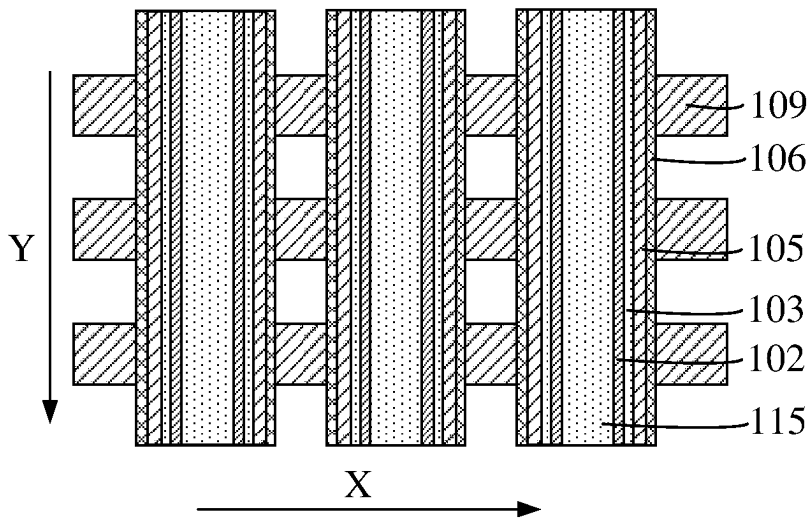
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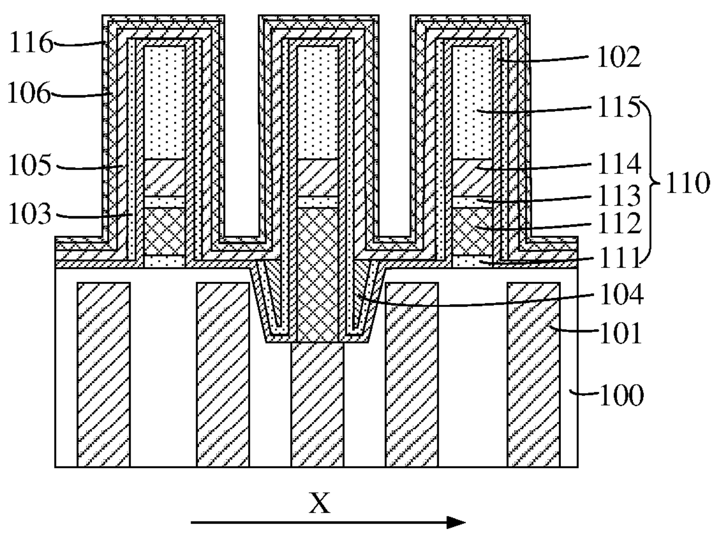
FIG. 14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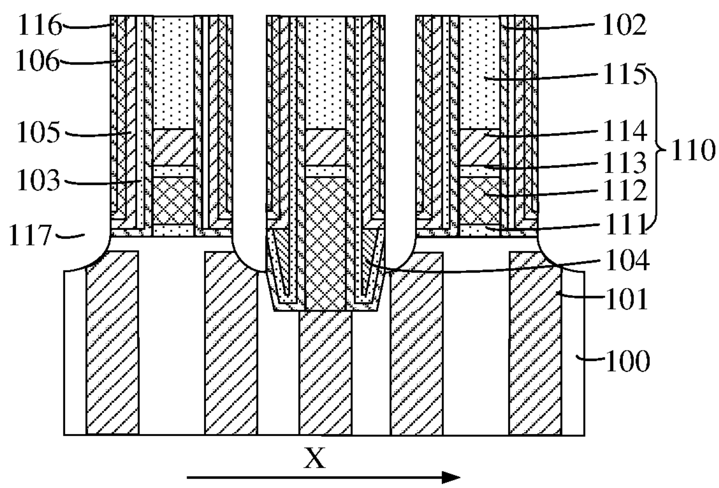
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FIG. 16

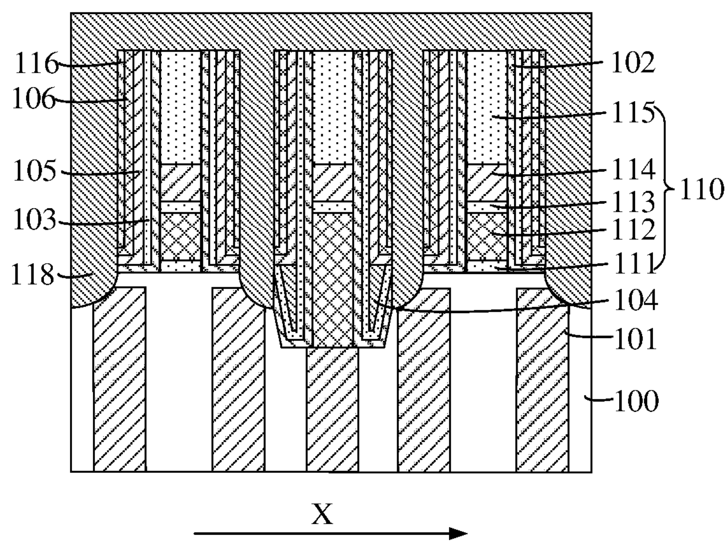
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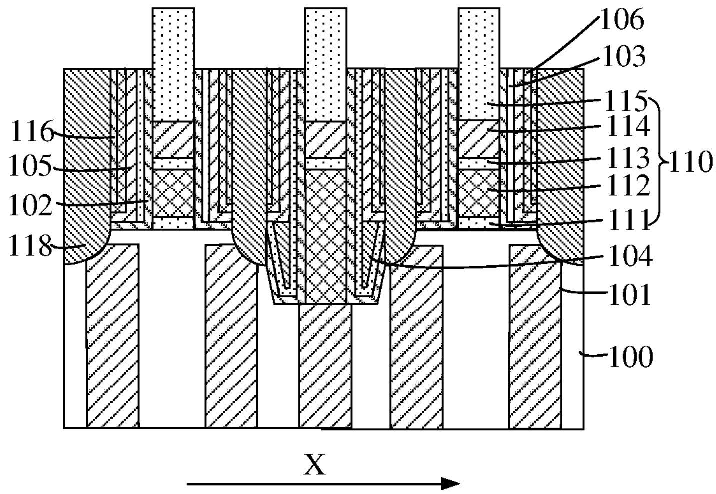
FIG. 17



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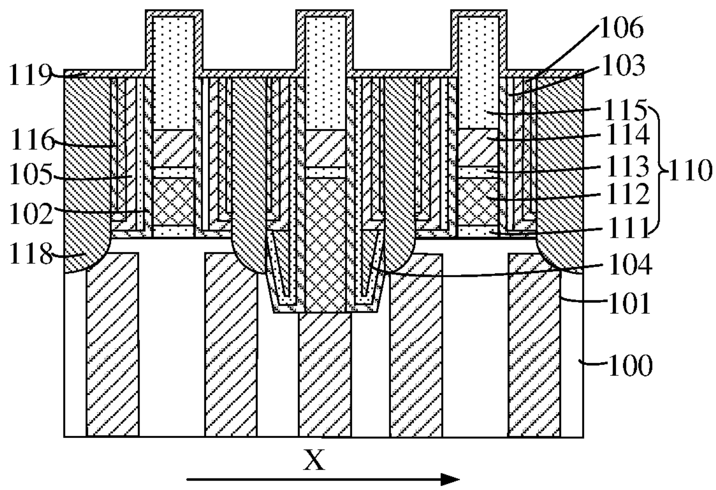

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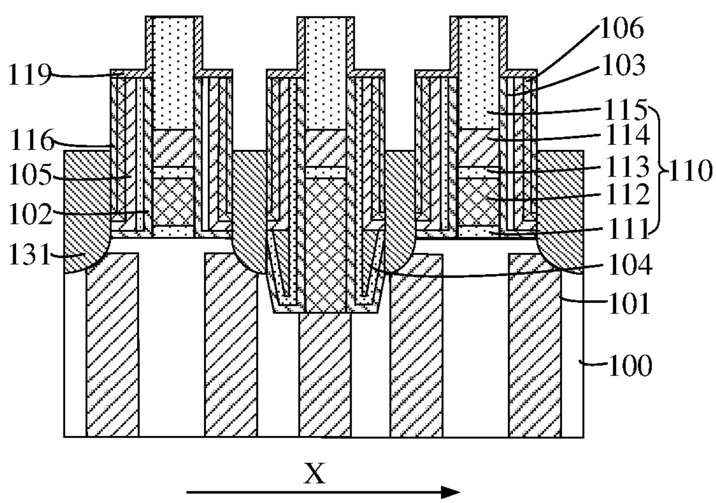

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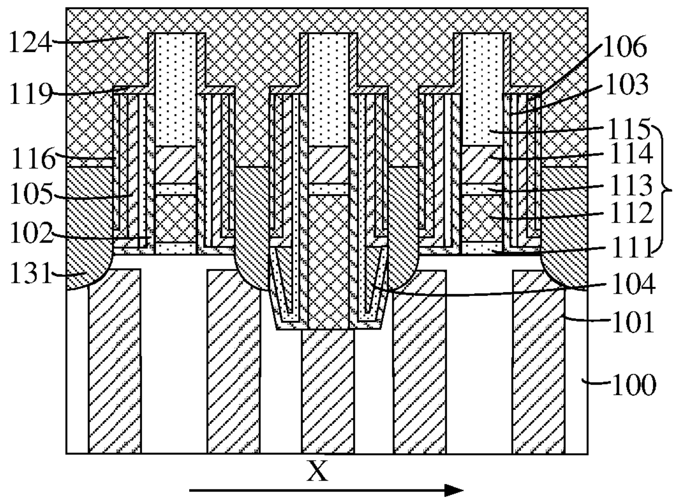
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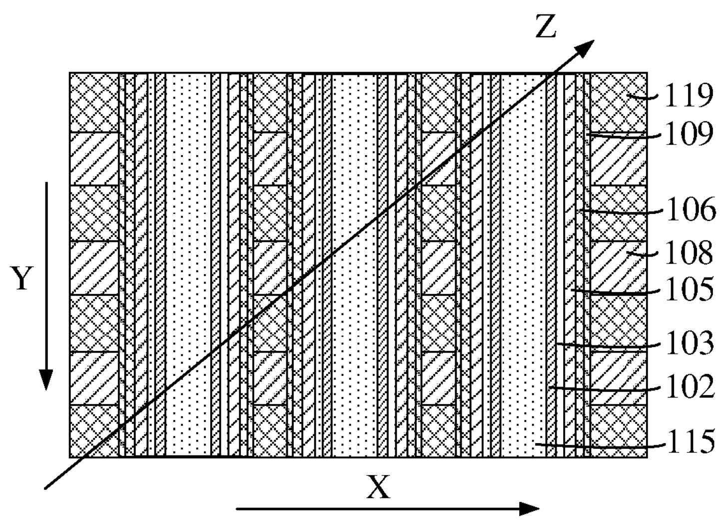
FIG. 21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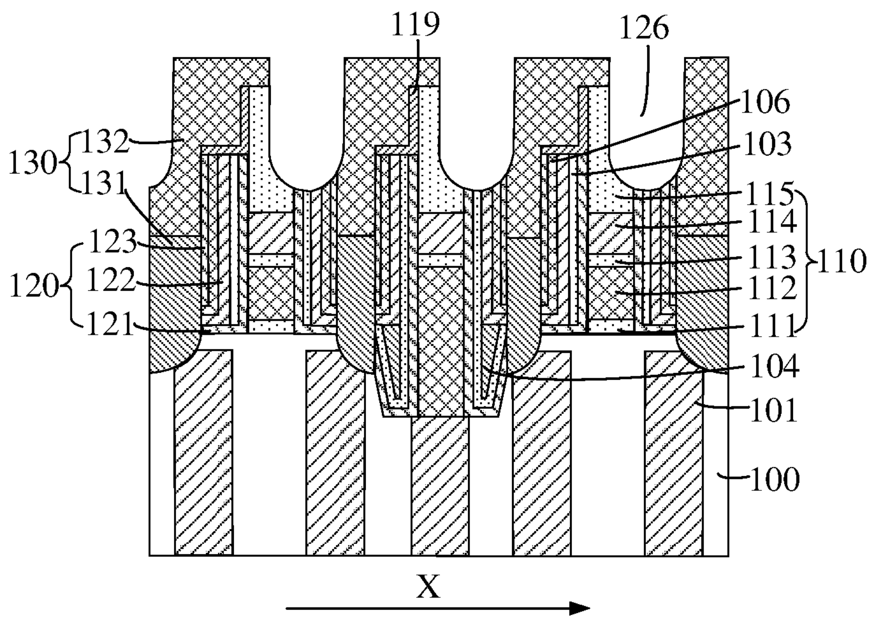
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FIG. 23

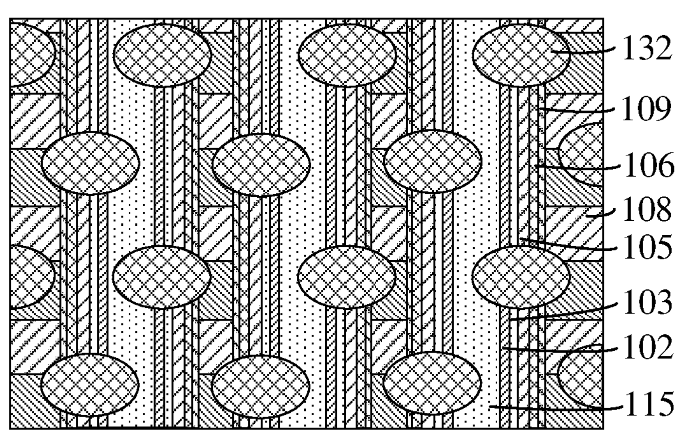
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FIG. 24

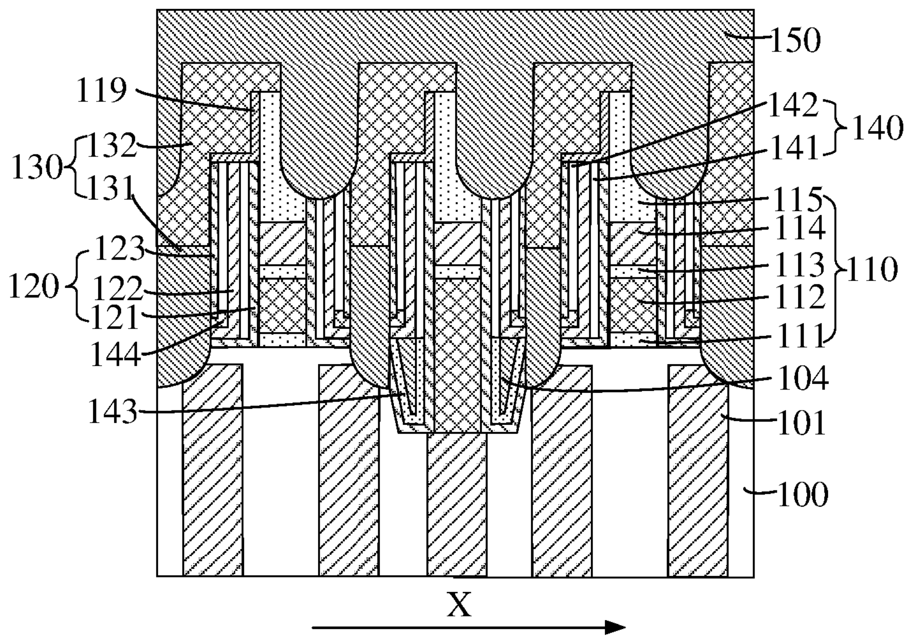
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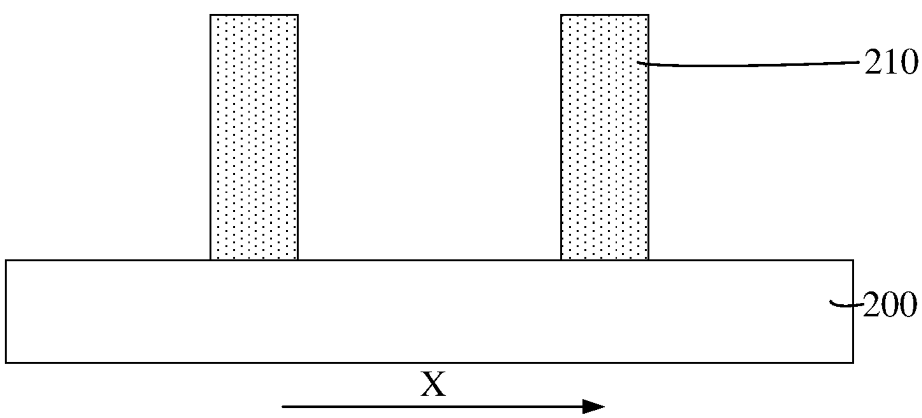
FIG. 25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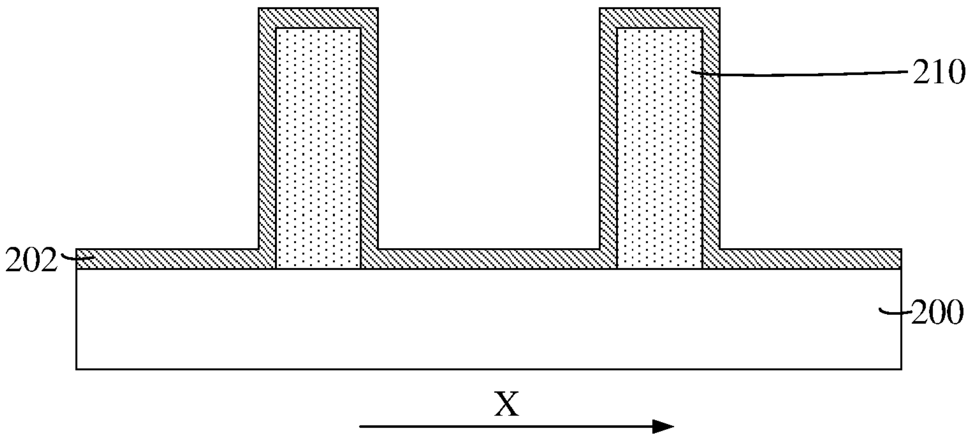

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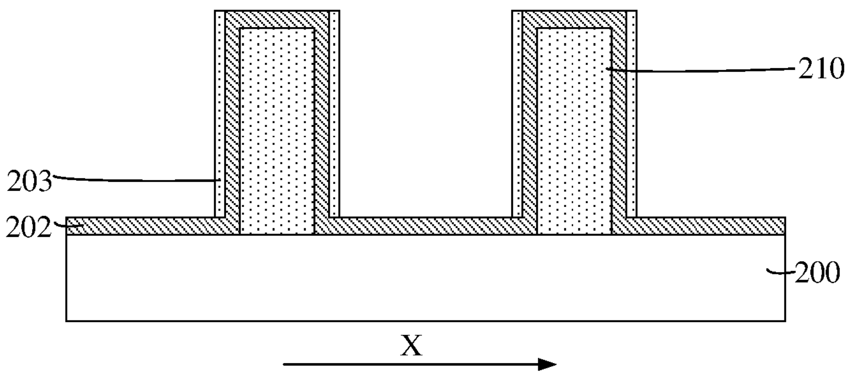

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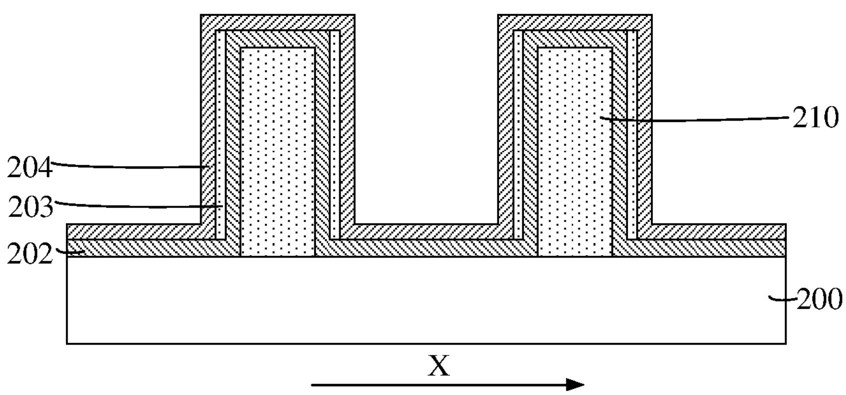

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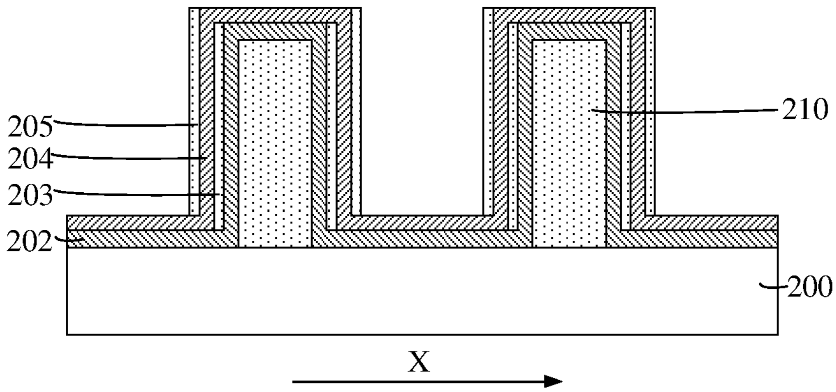


FIG. 30

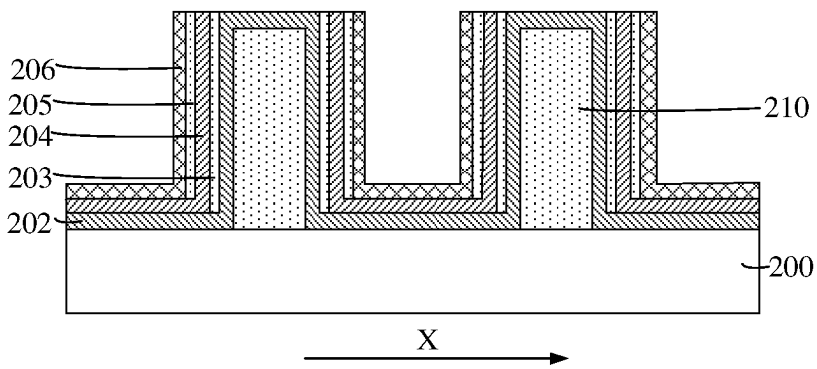
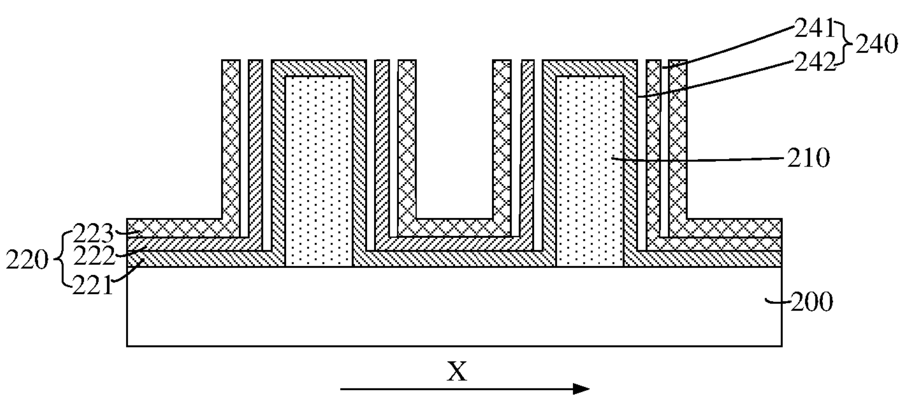


FIG. 31



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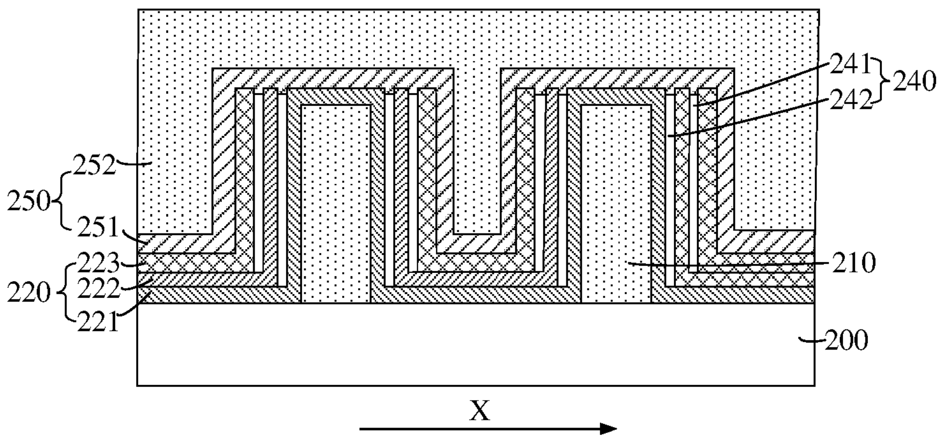


FIG. 33

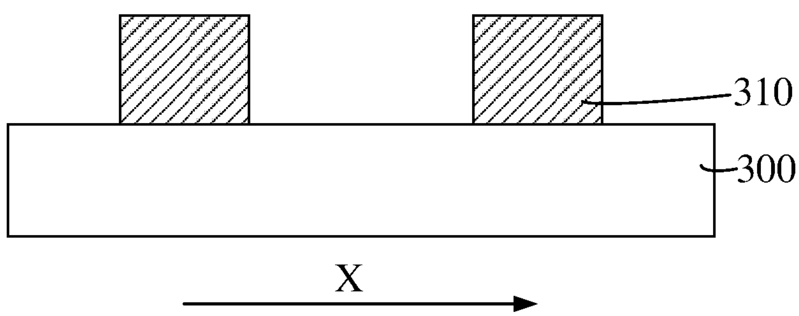
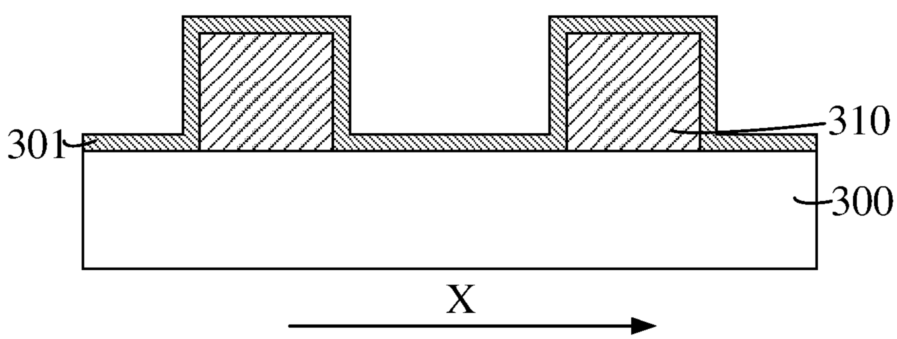


FIG. 34



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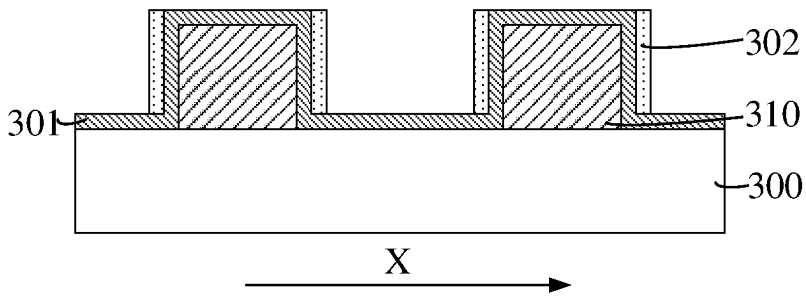

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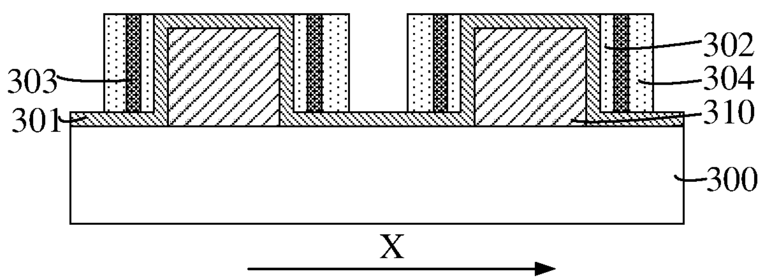

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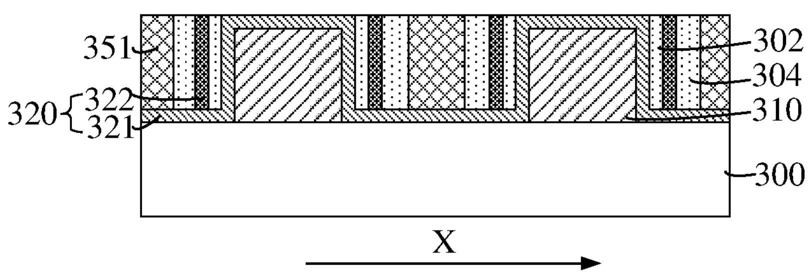

FIG. 38

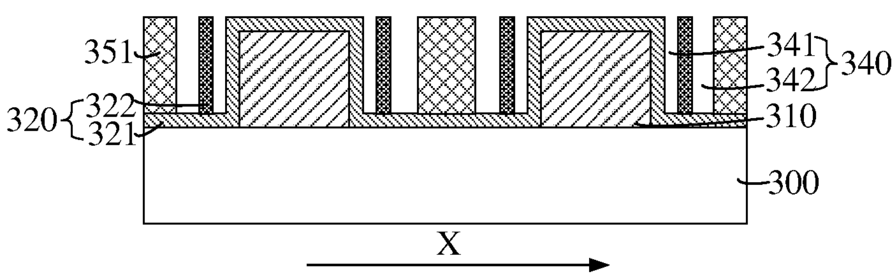

FIG. 39

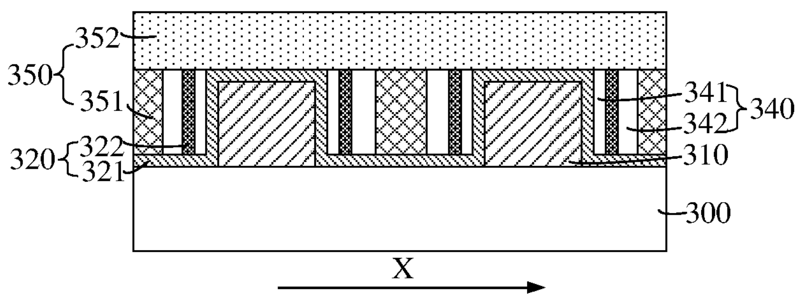

FIG. 40

SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority of Chinese Patent Application No. 202210266469.4, submitted to the Chinese Intellectual Property Office on Mar. 17, 2022, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

Embodiments of the present disclosure relate to the field of semiconductors, and in particular to, a semiconductor structure and a manufacturing method thereof.

BACKGROUND

With the continuous development of integrated circuit technology and process technology, to improve the integration degree of integrated circuits, the feature size of transistor (MOS) devices is continuously reduced. Under the process nodes such as the high-K metal gate (HKMG) and the fin transistor (Finfet), a series of problems need to be resolved while the operation speed of MOS devices is increased and its power consumption is reduced.

To reduce the power consumption of the MOS device, those skilled in the art use metal wires as the connection medium at a plurality of process nodes based on the advantage of the low resistance of the metal, which brings a series of problems. For example, a short circuit between adjacent metal wires leads to a chip failure, and a parasitic capacitance exists between adjacent metal wires.

SUMMARY

According to some embodiments of the present disclosure, an aspect of the embodiments of the present disclosure provides a semiconductor structure, including: a base; a plurality of first conductive structures, located on a surface of the base and distributed at intervals along a first direction; a plurality of second conductive structures, located on the surface of the base and the plurality of second conductive structures and the plurality of first conductive structures being arranged alternately; and a plurality of support structures, located on the surface of the base and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, where the given one of the plurality of support structures is provided with a support layer therein arranged along the first direction and a cavity divided by the support layer, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

According to some embodiments of the present disclosure, another aspect of the embodiments of the present disclosure provides a method of manufacturing a semiconductor structure, including: providing a base; forming a plurality of first conductive structures and a plurality of second conductive structures, where the plurality of first conductive structures are located on a surface of the base and are distributed at intervals along a first direction, and the plurality of second conductive structures are located on the surface of the base and the plurality of second conductive structures and the plurality of first conductive structures are arranged alternately; and forming a plurality of support structures located on the surface of the base and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, where the given one of the plurality of support structures is provided with a support layer therein arranged along the first direction and a cavity divided by the support layer, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

BRIEF DESCRIPTION OF THE DRAWINGS

One or more embodiments are exemplified by corresponding accompanying drawings, and these exemplified descriptions do not constitute a limitation on the embodiments. The accompanying drawings are not limited by scale unless otherwise specified. To describe the technical solutions in the embodiments of the present disclosure or in the prior art more clearly, the following outlines the accompanying drawings to be used in the embodiments of the present disclosure. Apparently, the accompanying drawings outlined below are merely some embodiments of the present disclosure. Those of ordinary skill in the art may derive other drawings from the outlined accompanying drawings without making any creative effort.

FIG. **1** is a schematic structural cross-sectional view of a semiconductor structure according to an embodiment of the present disclosure;

FIG. **2** is a schematic structural cross-sectional view of a semiconductor structure according to another embodiment of the present disclosure;

FIG. **3** is a schematic structural cross-sectional view of the semiconductor structure according to another embodiment of the present disclosure;

FIGS. **4** to **25** are schematic structural cross-sectional views corresponding to various steps of a method of manufacturing a semiconductor structure according to an embodiment of the present disclosure;

FIGS. **26** to **33** are schematic structural cross-sectional views corresponding to various steps of the method of manufacturing a semiconductor structure according to another embodiment of the present disclosure; and

FIGS. **34** to **40** are schematic structural cross-sectional views corresponding to various steps of the method of manufacturing a semiconductor structure according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

Embodiments of the present disclosure provide a semiconductor structure and a manufacturing method thereof. The semiconductor structure includes: a plurality of support structures located on the surface of the base, and a given one of the plurality of support structures being located between a given one of a plurality of first conductive structures and a given one of a plurality of second conductive structures. The given one of the plurality of support structures is provided with the support layer therein arranged along the first direction and a cavity divided by the support layer. The given one of the plurality of support structures extends along the side surface of the given one of the plurality of first conductive structures, and the cavity can reduce the parasitic capacitance of the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures. The support layer can function as a protection layer between the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures, which can avoid the short circuit caused by structural deformation at the sides of the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures. The cavity divided by the support layer and disposed in the given one of the plurality of support structures can avoid structural deformation due to various stresses in the subsequent process.

The embodiments of the present disclosure are described in detail below with reference to the accompanying drawings. However, those of ordinary skill in the art can understand that many technical details are proposed in the embodiments of the present disclosure to help readers better understand the present disclosure. However, even without these technical details and various changes and modifications made based on the following embodiments, the technical solutions claimed in the present disclosure may still be realized.

FIG. **1** is a schematic structural cross-sectional view of a semiconductor structure according to an embodiment of the present disclosure.

With reference to FIG. **1**, the semiconductor structure includes: a base **100**; a plurality of first conductive structures, located on a surface of the base **100** and distributed at intervals along a first direction X; a plurality of second conductive structures, located on the surface of the base **100**, and the plurality of second conductive structures and the plurality of first conductive structures being arranged alternately; and a plurality of support structures, located on the surface of the base **100** and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, where the given one of the plurality of support structures is provided with a support layer **120** therein arranged along the first direction X and a cavity **140** divided by the support layer, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

In some embodiments, the base **100** may be a made of a semiconductor material. The semiconductor material specifically may be any one selected from the group consisting of silicon, germanium, silicon germanide, or silicon carbide.

In some embodiments, the given one of the plurality of first conductive structures is a bit line **110** extending along the second direction. The second direction intersects the first direction X. Preferably, orthographic projection of the second direction and the orthographic projection of the first direction X on the base **100** are perpendicular to each other. The bit line **110** includes a first barrier layer **111**, a polysilicon layer **112**, a second barrier layer **113**, a conductive layer **114**, and a dielectric layer **115** that are stacked. The first barrier layer **111** is configured to prevent element diffusion between the polysilicon layer **112** and the base, and the second barrier layer **113** is used for ion diffusion between the conductive layer **114** and the polysilicon layer **112**, and reduce the contact resistance of a bit line contact hole. The material of the first barrier layer **111** may be one or more selected from the group consisting of silicon nitride, silicon oxide, or silicon carbide. The material of the polysilicon layer **112** may be one or more selected from the group consisting of amorphous silicon, microcrystalline silicon, or polycrystalline silicon. The material of the second barrier layer **113** may be TiN, SiN, titanium dinitride (Ti_2N_2), and titanium tetranitride (Ti_3N_4). The material of the conductive layer **114** may be metals such as tungsten, copper, and silver. The resistance of the metal is small, which is beneficial to improving the conductivity of the bit line **110** and the second conductive structure. The material of the dielectric layer **115** may be silicon oxide or silicon nitride.

In some embodiments, the given one of the plurality of second conductive structures is a storage node contact structure **130**, and the storage node contact structure **130** may specifically be a capacitor structure. The storage node contact structure **130** includes a capacitor contact layer **131** and a capacitor conductive layer **132**, and the capacitor contact layer **131** may be a polysilicon layer. The polysilicon layer may be doped with N-type doping elements or P-type doping elements. Doping elements may serve as carriers, to improve the migration and diffusion of carriers between the bit line **110** and the storage node contact structure **130**, thereby improving the electrical conductivity between the bit line **110** and the storage node contact structure **130**, and reducing the contact resistance of the storage node contact structure **130**. Specifically, the N-type element may be a group V element such as phosphorus (P), bismuth (Bi), antimony (Sb), or arsenic (As), and the P-type element may be a group III element such as boron (B), aluminum (Al), gallium (Ga) or indium (In). The capacitor conductive layer **132** is made of metals such as tungsten, copper, and silver, and the metal has a low resistance, which is therefore beneficial to improving the conductivity of the bit line **110** and the storage node contact structure **130**.

In some embodiments, in a direction perpendicular to the surface of the base **100**, the support layer **120** is as high as ⅓ to ⅔ of the bit line **110**, and the top surface of the support layer **120** away from the base **100** is not lower than that of the conductive layer **114**, which can contact between the bit line **110** and the storage node contact structure **130**, thereby avoiding the risk of a short circuit between the bit line **110** and the storage node contact structure **130**, and improving the safety of the semiconductor structure.

In some embodiments, a ratio of a thickness of the cavity **140** to that of the support layer **120** ranges from 1/3 to 4/3. This range set can avoid the risks caused by two aspects at the same time. On the one hand, if the ratio of the thickness of the cavity **140** to that of the support layer **120** is too small, that is, the thickness of the cavity **140** may be too small, the high-k constant (a K value) may be increased. Thus, there is a risk of an excessively large parasitic capacitance between the bit line **110** and the storage node contact structure **130**. On the other hand, if the foregoing ratio is too large, that is, the thickness of the support layer **120** may be too small, the support layer **120** may not be able to withstand various stresses, resulting in the deformation of the cavity **140**. Thus, an increased K value may cause the parasitic capacitance between the bit line **110** and the storage node contact structure **130** to increase, and lead to a short-circuit risk between them.

In some embodiments, the support layer **120** includes a first support layer **121**, a second support layer **122**, and a third support layer **123**, the first support layer **121** is located between the bit line **110** and the cavity **140**, the second support layer **122** is located in a middle of the cavity **140**, and the third support layer **123** is located between the storage node contact structure **130** and the cavity **140**; and a thickness of the first support layer **121** or the third support layer **123** is smaller than or equal to that of the second support layer **122**. The second support layer **122** in the cavity **140** is relatively thick, and can support the sides of the cavity **140**, such that the support structure can withstand various stresses in the subsequent process, thereby avoiding deformation of the side surfaces of the cavity **140**, and enabling the structure of the cavity **140** to be complete. This may avoid that the K value increases to cause the parasitic capacitance between the bit line **110** and the storage node contact structure **130** and lead to the short-circuit risk between them. The thickness of the second support layer **122** ranges from 2 nm to 4 nm, and may specifically be 2 nm, 2.3 nm, 3.1 nm, or 3.9 nm. The thickness of the first support layer **121** and the thickness of the third support layer **123** may both range from 1 nm to 2 nm, and specifically may be 1 nm, 1.3 nm, 1.6 nm, or 1.9 nm.

It can be understood that the first support layer **121** and the third support layer **123** may have a same thickness or not. Their thicknesses may be set respectively according to performances of the adjacent first conductive structure and the adjacent second conductive structure.

In some embodiments, the first support layer **121**, the second support layer **122**, and the third support layer **123** may be made of a same material such as silicon boron carbon nitride. Because silicon boron carbon nitride is relatively strong and hard, the three support layers made of it are also relatively strong and hard, and can withstand various stresses in the subsequent process, thereby avoiding deformation of the side surfaces of the cavity **140**, and enabling the structure of the cavity **140** to be complete. This may avoid that the K value increases to cause the parasitic capacitance between the bit line **110** and the storage node contact structure **130** to increase and lead to the short-circuit risk between them. In some other embodiments, the first support layer **121**, the second support layer **122**, and the third support layer **123** may be made of different materials such as silicon nitride or silicon carbonitride.

In some embodiments, the cavity **140** includes a first cavity **141** and a second cavity **142** separated by the second support layer **122**. The thickness of the first cavity **141** and the thickness of the second cavity **142** both range from 1 nm to 3 nm, and may be specifically, 1 nm, 1.3 nm, 2.2 nm, or 2.9 nm. While ensuring that the parasitic capacitance between the bit line **110** and the storage node contact structure **130** is small, the thickness range of the first cavity **141** and the second cavity **142** ensures that the thickness of the support layer **120** is large. The support layer **120** with such a thickness can withstand various stresses in the subsequent process, to avoid structural deformation of the first cavity **141** and the second cavity **142**. It can be understood that the first cavity **141** and the second cavity **142** can have a same thickness or not.

In some embodiments, a first isolation film **143** is provided at the bottom of the first cavity **141** close to the base **100**. A second isolation film **144** is provided at the bottom of the second cavity **142** close to the base **100**. Along the direction perpendicular to the base **100**, the top surface of the first isolation film **143** away from the base **100** is not higher than the conductive layer **114** of the bit line **110**, and the top surface of the second isolation film **144** away from the base **100** is not higher than the conductive layer **114** of the bit line **110**. In this way, there is enough space in the first cavity **141** and the second cavity **142** to decrease the K value, which can reduce the parasitic capacitance between the bit line **110** and the storage node contact structure **130**. Specifically, a thickness of the first isolation film **143** ranges from 0 nm to 10 nm, such as 1 nm, 3 nm, 5.6 nm, or 8.9 nm. The thickness of the second isolation film **144** ranges from 0 nm to 10 nm, such as 1.3 nm, 3.5 nm, 6.6 nm, or 9.8 nm.

In some embodiments, the semiconductor structure further includes: a protective layer **119**, an insulating layer **104**, and an isolation structure **150**. The protective layer **119** is located between the given one of the plurality of support structures and the given one of the plurality of storage node contact structures **130**. The insulating layer **104** is located in the plurality of support structures, and the insulating layer **104** is located between the first isolation film **143** and the second support layer **122**. The isolation structure **150** is located between the given one of the plurality of bit lines **110** and the given one of the plurality of storage node contact structures **130** that are adjacent to each other. The isolation structure **150** is located on the top surfaces of plurality of the bit lines **110** and the top surfaces of the plurality of storage node contact structures **130**. The material of the protective layer **119** may be silicon boron carbon nitride, silicon nitride, or silicon carbonitride, and the material of the insulating layer **104** may be silicon dioxide, silicon carbide, or silicon nitride. The isolation structure **150** may be a single-layer structure or a stacked multi-layer structure, and the material of the isolation structure **150** may be one or more selected from the group consisting of silicon dioxide, silicon carbide, or silicon nitride.

In the technical solutions provided by the embodiment (FIG. **1**) of the present application, the given one of the plurality of support structures is located on the surface of the base **100** and between the given one of the plurality of bit lines **110** and the given one of the plurality of storage node contact structures **130**. The given one of the plurality of support structures is provided with the support layer **120** therein arranged along the first direction X and a cavity **140** divided by the support layer **120**. The given one of the plurality of support structures extends along the side surface of the given one of the plurality of bit lines **110**, and the cavity **140** can reduce the parasitic capacitance of the given one of the plurality of bit lines **110** and the given one of the plurality of storage node contact structures **130**. The support layer **120** can function as a protection layer between the given one of the plurality of bit lines **110** and the given one of the plurality of storage node contact structures **130**, which can avoid the short circuit caused by structural deformation at the sides of the given one of the plurality of bit lines **110** and the given one of the plurality of storage node contact structures **130**. The cavity **140** divided by the support layer **120** and disposed in the given one of the plurality of support structures can avoid structural deformation due to various stresses in the subsequent process.

Correspondingly, another embodiment of the present disclosure provides a semiconductor structure. A semiconductor structure provided by another embodiment of the present disclosure is substantially the same as the one provided by the foregoing embodiment (FIG. **1**). The main difference is that the given one of the plurality of first conductive structures provided by the foregoing embodiment of the present disclosure is a bit line, and the given one of the plurality of second conductive structures is a storage node contact structure, and the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures in the semiconductor structure provided by the another embodiment of the present disclosure are both gate structures. Details of contents or elements the same as or similar to those described in the foregoing embodiments are not repeated, and the ones different from the above are described in detail. The semiconductor structure provided by the another embodiment of the present disclosure is described in detail below with reference to FIG. **2**.

FIG. **2** is a schematic structural cross-sectional view of a semiconductor structure according to another embodiment of the present disclosure.

With reference to FIG. 2, the semiconductor structure includes: a base **200**; a plurality of first conductive structures located on a surface of the base **200** and distributed at intervals along a first direction X; a plurality of second conductive structures located on the surface of the base **200**, and the plurality of second conductive structures and the plurality of first conductive structures being arranged alternately; a plurality of support structures located on the surface of the base **200** and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, where the given one of the plurality of support structures is provided with a support layer **220** therein arranged along the first direction X and a cavity **240** divided by the support layer **220**, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

In some embodiments, the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures are both the gate structures **210**, and the gate structures **210** serve as the word lines of the semiconductor structure. The plurality of gate structures **210** are spaced apart along the first direction X, and the given one of the plurality of support structures are located between adjacent gate structures **210**.

In some embodiments, along the first direction X, the support layer **220** includes a first spacing layer **221**, a second spacing layer **222**, and a third spacing layer **223**, the first spacing layer **221** is located between the gate structure **210** and the cavity **240**, the second spacing layer **222** is located in a middle of the cavity **240**, and the third spacing layer **223** is located between the isolation structure **250** and the cavity **240**; and a thickness of the first spacing layer **221** or the third spacing layer **223** is smaller than or equal to that of the second spacing layer **222**. The second spacing layer **222** in the cavity **240** is relatively thick, and can support the side surface of the cavity **240**, such that the support structure can withstand various stresses in the subsequent process, thereby avoiding deformation of the side surfaces of the cavity **240**, and enabling the structure of the cavity **240** to be complete. This may avoid that the K value increases to cause the parasitic capacitance between adjacent gate structures **210** to increase and lead to the short-circuit risk between them. The thickness of the first spacing layer **221** ranges from 2 nm to 8 nm, and may specifically be 2 nm, 4.6 nm, 6.8 nm, or 7.8 nm. The thickness of the second spacing layer **222** may range from 2 nm to 10 nm, and may specifically be 2 nm, 5.6 nm, 8.1 nm, or 9.8 nm. The thickness of the third spacing layer **223** may range from 4 nm to 10 nm, and may specifically be 4 nm, 5.3 nm, 8.6 nm, or 9.9 nm.

In some embodiments, the first spacing layer **221**, the second spacing layer **222**, and the third spacing layer **223** may be made of a same material such as silicon boron carbon nitride. Because silicon boron carbon nitride is relatively strong and hard, the three spacing layers made of it are also relatively strong and hard, and can withstand various stresses in the subsequent process, thereby avoiding deformation of the side surfaces of the cavity **140**, and enabling the structure of the cavity **140** to be complete. This may avoid that the K value increases to cause the parasitic capacitance between adjacent gate structures **210** to increase and lead to the short-circuit risk between them. In some other embodiments, the first spacing layer **221**, the second spacing layer **222**, and the third spacing layer **223** may be made of different materials such as silicon nitride or silicon carbonitride.

In some embodiments, along the first direction X, the thickness of the cavity **240** ranges from 2 nm to 20 nm. The cavity **240** includes a first cavity **241** and a second cavity **242** separated by the second spacing layer **222**. The thickness of the first cavity **241** and the thickness of the second cavity **242** both range from 2 nm to 10 nm, and may be specifically, 2 nm, 3.8 nm, 6.2 nm, or 8.9 nm. While ensuring that the parasitic capacitance between adjacent gate structures **210** is small, the thickness range of the first cavity **241** and the second cavity **242** ensures that the support layer **220** is relatively thick. The support layer **220** with such a thickness can withstand various stresses in the subsequent process, to avoid structural deformation of the first cavity **241** and the second cavity **242**. It can be understood that the first cavity **241** and the second cavity **242** can have a same thickness or not.

In some embodiments, the semiconductor structure further includes an isolation structure **250** located between adjacent two of the plurality of support structures, where the given one of the plurality of support structures also extends along a side surface of the given one of the plurality of second conductive structures. Specifically, the given one of the plurality of support structures extends along the side surface of the given one of the plurality of gate structures **210**. The isolation structure **250** includes a first isolation layer **251** and a second isolation layer **252** that are stacked. The materials of the first isolation layer **251** and the second isolation layer **252** may be one or more selected from the group consisting of silicon dioxide, silicon carbide, or silicon nitride.

In the technical solutions provided by the embodiment (FIG. **2**) of the present application, the given one of the plurality of support structures is located on the surface of the base **200** and between adjacent gate structures **210**. The given one of the plurality of support structures is provided with the support layer **220** therein arranged along the first direction X and a cavity **240** divided by the support layer **220**. The given one of the plurality of support structures extends along the side surface of the given one of the plurality of gate structures **210**, and the cavity **240** can reduce the parasitic capacitance between adjacent gate structures **210**. The support layer **220** can function as a protection layer between adjacent gate structures **210**, which can avoid the short circuit caused by structural deformation at the sides of adjacent gate structures **210**. The cavity **240** separated by the support layer **220** and disposed in the given one of the plurality of support structures can avoid structural deformation due to various stresses in the subsequent process.

Correspondingly, another embodiment of the present disclosure also provides a semiconductor structure. A semiconductor structure provided by another embodiment of the present disclosure is substantially the same as the one provided by the foregoing embodiment (FIG. **1** or **2**). The main difference is that the given one of the plurality of first conductive structures provided by the foregoing embodiment of the present disclosure is a bit line, and the given one of the plurality of second conductive structures is a storage node contact structure, and the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures in the semiconductor structure provided by the another embodiment of the present disclosure are both metal wire structures. Details of contents or elements the same as or similar to those described in the foregoing embodiments are not repeated, and the ones different from the above are described in detail. The semiconductor structure provided by the another embodiment of the present disclosure is described in detail below with reference to FIG. **3**.

FIG. **3** is a schematic structural cross-sectional view of the semiconductor structure according to another embodiment of the present disclosure.

With reference to FIG. **3**, the semiconductor structure includes: a base **300**; a plurality of first conductive structures located on a surface of the base **300** and distributed at intervals along a first direction X; a plurality of second conductive structures located on the surface of the base **300**, and the plurality of second conductive structures and the plurality of first conductive structures being arranged alternately; a plurality of support structures located on the surface of the base **300** and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, where the given one of the plurality of support structures is provided with a support layer **320** therein arranged along the first direction X and a cavity **340** divided by the support layer **320**, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

In some embodiments, the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures are both metal wire structures **310**, and the metal wire structures **310** serve as the solder joints of the semiconductor structure. The plurality of metal wire structures **310** are spaced apart along the first direction X, and the support structures are located between adjacent metal wire structures **310**.

In some embodiments, along the first direction X, the support layer **320** includes a fourth spacing layer **321** and a fifth spacing layer **322**. The fourth spacing layer **321** is located between the metal wire structure **310** and the cavity **340**. The fifth spacing layer **322** is located in the middle of the cavity **340**. The fourth spacing layer **321** is thinner than the fifth spacing layer **322**. The fifth spacing layer **322** in the cavity **340** is relatively thick, and can support the side surface of the cavity **340**, such that the support structure can withstand various stresses in the subsequent process, thereby avoiding deformation of the side surfaces of the cavity **340**, and enabling the structure of the cavity **340** to be complete. This may avoid that the K value increases to cause the parasitic capacitance between adjacent metal wire structures **310** to increase and lead to the short-circuit risk between them. The thickness of the fourth spacing layer **321** may range from 2 nm to 8 nm, and may specifically be 2 nm, 4.6 nm, 6.8 nm, or 7.8 nm. The thickness of the fifth spacing layer **322** may range from 4 nm to 10 nm, and may specifically be 4 nm, 5.3 nm, 8.6 nm, or 9.9 nm.

In some embodiments, along the first direction X, the thickness of the cavity **340** ranges from 2 nm to 20 nm. The cavity **340** includes a first cavity **341** and a second cavity **342** separated by the fourth spacing layer **322**. The thickness of the first cavity **341** and the thickness of the second cavity **342** both range from 2 nm to 10 nm, and may be specifically, 2 nm, 3.8 nm, 6.2 nm, or 8.9 nm.

In some embodiments, the semiconductor structure further includes an isolation structure **350** located between adjacent two of the plurality of support structures, where the given one of the plurality of support structures also extends along a side surface of the given one of the plurality of second conductive structures. Specifically, the given one of the plurality of support structures extends along the side surface of the given one of the plurality of metal wire structures **310**. Along the first direction X, the thickness of the isolation structure **350** ranges from 5 nm to 30 nm. The isolation structure **350** includes a first isolation layer **351** and a second isolation layer **352**. The first isolation layer **351** is located between adjacent support structures, and the second isolation layer **352** is located on the top surfaces of the support structures, the top surfaces of the metal wire structures **310**, and the top surface of the first isolation layer **351**. Materials of the first isolation layer **351** and the second isolation layer **352** may be one or more selected from the group consisting of silicon dioxide, silicon carbide, or silicon nitride.

In the technical solutions provided by the embodiment (FIG. **3**) of the present application, the given one of the plurality of support structures is located on the surface of the base **300** and between adjacent metal wire structures **310**. The given one of the plurality of support structures is provided with the support layer **320** therein arranged along the first direction X and a cavity **340** divided by the support layer **320**. The given one of the plurality of support structures extends along the side surface of the given one of the plurality of metal wire structures **310**, and the cavity **340** can reduce the parasitic capacitance between adjacent metal wire structures **310**. The support layer **320** can function as a protection layer between adjacent metal wire structures **310**, which can avoid the short circuit caused by structural deformation at the sides of adjacent metal wire structures **310**. The cavity **340** separated by the support layer **320** and disposed in the given one of the plurality of support structures can avoid structural deformation due to various stresses in the subsequent process.

Some embodiments of the present disclosure provide a method of manufacturing a semiconductor structure, which can be used to form the semiconductor structure provided by some foregoing embodiments. The method of manufacturing a semiconductor structure provided by some embodiments of the present disclosure will be described below in detail with reference to the accompanying drawings. FIGS. **4** to **25** are schematic structural cross-sectional views corresponding to various steps of a method of manufacturing a semiconductor structure according to an embodiment of the present disclosure. FIGS. **11** to **14**, **22**, and **24** are schematic structural cross-sectional views of a semiconductor structure taken along A-A.

With reference to FIGS. **4** to **25**, the method of manufacturing a semiconductor structure includes: providing a base **100**; forming a plurality of first conductive structures and a plurality of second conductive structures, where the plurality of first conductive structures are located on a surface of the base **100** and distributed at intervals along a first direction X, and plurality of second conductive structures located on the surface of the base **100**, and the plurality of second conductive structures and the plurality of first conductive structures are arranged alternately; and forming a plurality of support structures located on the surface of the base **100** and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, where the given one of the plurality of support structures is provided with a support layer **120** therein arranged along the first direction X and a cavity **140** divided by the support layer **120**, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

With reference to FIG. **4**, a base **100** is provided, a plurality of active structures **101** are arranged at intervals in the base, and a plurality of bit lines **110** are formed. The plurality of bit lines **110** are located on the surface of the base and a given one of the plurality of bit lines **110** is in contact with a given one of the plurality of active structures **101**.

In some embodiments, the given one of the plurality of first conductive structures is a bit line **110** extending along the second direction. The second direction intersects the first direction X. Preferably, orthographic projection of the second direction and the orthographic projection of the first direction X on the base **100** are perpendicular to each other. The bit line **110** includes a first barrier layer **111**, a polysilicon layer **112**, a second barrier layer **113**, a conductive layer **114**, and a dielectric layer **115** that are stacked.

With reference to FIGS. **5** to **24**, the step of forming the plurality of support structures includes: sequentially forming a first support film **102**, a first oxide film **103**, a second support film **105**, a second oxide film **106**, and a third support film **116** on the base **100** and on top surfaces and side surfaces of the plurality of first conductive structures; removing, through etching, the first support film **102**, the first oxide film **103**, the second support film **105**, the second oxide film **106**, and the third support film **116** by a thickness, to expose the plurality of first conductive structures with a thickness; and removing the first oxide film **103** and the second oxide film **106** through etching, to form the cavity **140**, where remaining parts of the first support film **102**, second support film **105**, and third support film **116** form the support layer.

In some embodiments, the support layer **120** includes a first support layer **121**, a second support layer **122**, and a third support layer **123**, the first support layer **121** is located between the bit line **110** and the cavity **140**, the second support layer **122** is located in a middle of the cavity **140**, and the third support layer **123** is located between the storage node contact structure formed subsequently and the cavity **140**; and a thickness of the first support layer **121** or the third support layer **123** is smaller than or equal to that of the second support layer **122**. The second support layer **122** in the cavity **140** is relatively thick, and can support the sides of the cavity **140**, such that the support structure can withstand various stresses in the subsequent process, thereby avoiding deformation of the side surfaces of the cavity **140**, and enabling the structure of the cavity **140** to be complete. This may avoid that the K value increases to cause the parasitic capacitance between the bit line **110** and the storage node contact structure formed subsequently and lead to the short-circuit risk between them.

With reference to FIG. **5**, a first support film **102** is formed on the surface of the base **100** and the top surfaces and side surfaces of the bit lines **110**. In some embodiments, the first support film **102** is formed through the epitaxial growth process. A thickness of the first support film **102** ranges from 1 to 2 nm, and may specifically be 1 nm, 1.3 nm, 1.6 nm or 1.9 nm. The material of the first support film **102** may be silicon boron carbon nitride, silicon nitride, or silicon carbonitride.

With reference to FIG. **6**, a first oxide film **103** is formed on the surface of the first support film **102**. In some embodiments, the first oxide film **103** is formed through the atomic layer deposition process. A thickness of the first oxide film **103** ranges from 1 to 3 nm, and may specifically be 1 nm, 1.3 nm, 2.3 nm, or 2.8 nm. The material of the first oxide film **103** may be silicon dioxide or silicon carbonitride.

In some embodiments, an etch selectivity between a material of the first oxide film **103** and a material of the first support film **102** is greater than 50.

With reference to FIG. **7**, an insulating layer **104** is formed on the surface of the first oxide film **103**, and a part of the insulating layer **104** higher than the top surface of the first oxide film **103** is removed through etching. In some embodiments, the insulating layer **104** is formed through a thin film deposition process, and a part of the insulating layer **104** higher than the top surface of the first oxide film **103** is removed through a wet etching process. The material of the insulating layer **104** is silicon dioxide or silicon carbonitride.

With reference to FIG. **8**, of the first oxide film **103**, a part higher than the top surface of the first support film **102** and a part at the bottom are removed through etching, and a part of the first oxide film **103** on the side surfaces of the bit lines **110** is retained. In some embodiments, a dry etching process is used to remove a part of the first oxide film **103** to expose the surface of the first support film **102**, which is beneficial for the structural stability of the support layer formed subsequently.

With reference to FIG. **9**, a second support film **105** and a second oxide film **106** are sequentially formed on the surface of the first support film **102** and the first oxide film **103**. In some embodiments, the second support film **105** and the second oxide film **106** are formed through the atomic layer deposition process. A thickness of the second support film **105** ranges from 2 nm to 4 nm, and may specifically be 2 nm, 2.3 nm, 3.1 nm, or 3.9 nm. The second support film **105** and the first support film **102** may have a same material or not. The second oxide film **106** and the first oxide film **103** may have a same thickness or not. The second oxide film **106** and the first oxide film **103** may have a same material or not.

In some embodiments, an etch selectivity between a material of the first oxide film **103** and a material of the second support film **105** is greater than 50. An etch selectivity between a material of the second oxide film **106** and a material of the first support film **102** is greater than 50. An etch selectivity between the material of the second oxide film **106** and the material of the second support film **105** is greater than 50.

With reference to FIGS. **10** to **14**, a spin-on dielectric (SOD) structure is formed to backfill gaps between bit lines, thereby forming a capacitor structure. The SOD structure isolates the transistors in the semiconductor.

With reference to FIGS. **10** and **11**, a sacrificial film **107** is formed between adjacent bit lines **110**, and on the top surface of the second oxide film **106**. The sacrificial film **107** is formed through a chemical vapor deposition (CVD) process.

Specifically, with reference to FIG. **12**, along the second direction Y, the sacrificial film **107** is etched to form the sacrificial layers **108** arranged at intervals. With reference to FIG. **13**, an isolation layer **109** is formed to fill up the region between adjacent sacrificial layers **108**. With reference to FIG. **14**, the sacrificial layer is removed, to expose the base **100** or the second oxide film **106** at the bottom.

With reference to FIG. **15**, a third support film **116** is formed on the surface of the second oxide film **106**. In some embodiments, the third support film **116** is formed through the atomic layer deposition process. The third support film **116** and the first support film **102** may have a same thickness or not. The third support film **116** and the first support film **102** may be made of a same material or not.

With reference to FIG. **16**, a first through hole **117** is formed, and the bottom of the first through hole **117** exposes the surface of the active structure **101** in the base **100**, such that the subsequently formed storage node contact structure is connected to the active structure **101**.

With reference to FIG. **17**, a polysilicon film **118** is formed on the surface of the third support film **116** to fill up the first through hole **117**.

With reference to FIG. **18**, the polysilicon film **118** is etched for the first time, while the first support film **102**, the first oxide film **103**, the second support film **105**, the second oxide film **106**, and the third support film **116** are etched by a thickness. In some embodiments, the polysilicon film **118** is etched for the first time through dry etching or wet etching.

With reference to FIG. **19**, a protective layer **119** is formed on the top surfaces of the first support film **102**, the first oxide film **103**, the second support film **105**, the second oxide film **106**, the third support film **116**, and the bit lines **110**, to protect the bit lines **110**. The material of the protective layer **119** may be silicon boron carbon nitride, silicon nitride, or silicon carbonitride.

With reference to FIG. **20**, the polysilicon film **118** is etched for the second time to form a polysilicon layer, while the protective layer **119** on the tops of the bit lines **110** is etched. The polysilicon layer is the capacitor contact structure **131**.

With reference to FIG. **21**, a conductive film **124** is formed on the top surfaces of the capacitor contact structures **131** and the bit lines **110**.

With reference to FIGS. **22** to **24**, the conductive film **124** is etched in the first direction X and the third direction Z to form the capacitive conductive layers **132** arranged in a staggered or hexagonal manner. The capacitor contact layer **131** and the capacitor conductive layer **132** form a storage node contact structure **130**, and the storage node contact structure **130** serves as a second conductive structure.

With reference to FIG. **25**, the isolation structure **150** is formed. The isolation structure **150** is located between the bit line **110** and the storage node contact structure **130** that are adjacent, and the isolation structure **150** is located on the top surfaces of the bit lines **110** and the top surfaces of the storage node contact structures **130**. The isolation structure **150** may be a single-layer structure or a stacked multi-layer structure, and the material of the isolation structure **150** may be one or more selected from the group consisting of silicon dioxide, silicon carbide, or silicon nitride.

Correspondingly, another embodiment of the present disclosure provides a method of manufacturing a semiconductor structure. A method of manufacturing a semiconductor structure provided by another embodiment of the present disclosure is substantially the same as the ones provided by the foregoing embodiments (FIGS. **4** to **25**). The main difference is that the given one of the plurality of first conductive structures provided by the foregoing embodiments of the present disclosure is a bit line, and the given one of the plurality of second conductive structures is a storage node contact structure, and the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures in the semiconductor structure provided by the another embodiment of the present disclosure are both gate structures. Details of contents or elements the same as or similar to those described in the foregoing embodiments are not repeated, and the ones different from the above are described in detail. The method of manufacturing a semiconductor structure provided by another embodiment of the present disclosure is described in detail below with reference to FIGS. **26** to **33**.

FIGS. **26** to **33** are schematic structural cross-sectional views corresponding to various steps of the method of manufacturing a semiconductor structure according to another embodiment of the present disclosure.

With reference to FIGS. **26** to **33**, the method of manufacturing a semiconductor structure includes: providing a base **200**; forming a plurality of first conductive structures and a plurality of second conductive structures, where the plurality of first conductive structures are located on a surface of the base **200** and distributed at intervals along a first direction X, and the plurality of second conductive structures are located on the surface of the base **200**, and the plurality of second conductive structures and the plurality of first conductive structures are arranged alternately; and forming a plurality of support structures located on the surface of the base **200** and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, where the given one of the plurality of support structures is provided with a support layer **220** therein arranged along the first direction X and a cavity **240** divided by the support layer **220**, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

With reference to FIG. **26**, a base **200** is provided and a plurality of gate structures **210** are formed on the surface of the base **200** and arranged at intervals along the first direction X.

The given one of the plurality of first conductive structures and the given one of the plurality of plurality of second conductive structures are both the gate structures **210**, and the gate structures **210** serve as the word lines of the semiconductor structure. The support structures are located between adjacent gate structures **210**.

With reference to FIGS. **27** to **32**, the step of forming the plurality of support structures includes: sequentially forming a first spacing film **202**, a third oxide film **203**, a second spacing film **204**, a fourth oxide film **205**, and a third spacing film **206** on the surface of the base **200** as well as top surfaces and side surfaces of the gate structures **210**; removing, through etching, a part of the third oxide film **203**, a part of the second spacing film **204**, a part of the fourth oxide film **205**, and a part of the third spacing film **206** that are higher than a top of the first spacing film **202**; and removing the third oxide film **203** and the fourth oxide film **205** through etching to form the cavity **240**, where remaining parts of the first spacing film **202**, second spacing film **204**, and third spacing film **206** form the support layer **220**.

In some embodiments, along the first direction X, the support layer **220** includes a first spacing layer **221**, a second spacing layer **222**, and a third spacing layer **223**, the first spacing layer **221** is located between the gate structure **210** and the cavity **240**, the second spacing layer **222** is located in a middle of the cavity **240**, and the third spacing layer **223** is located between the isolation structure **250** and the cavity **240**; and a thickness of the first spacing layer **221** or the third spacing layer **223** is smaller than or equal to that of the second spacing layer **222**. The second spacing layer **222** in the cavity **240** is relatively thick, and can support the side surface of the cavity **240**, such that the support structure can withstand various stresses in the subsequent process, thereby avoiding deformation of the side surfaces of the cavity **240**, and enabling the structure of the cavity **240** to be complete. This may avoid that the K value increases to cause the parasitic capacitance between adjacent gate structures **210** to increase and lead to the short-circuit risk between them.

With reference to FIG. **27**, a first spacing film **202** is formed on the surface of the base **200** and the top surface and side surface of the gate structure **210**. In some embodiments, the first support film **102** is formed through the epitaxial growth process. A thickness of the first spacing film **202** ranges from 2 to 8 nm, and may specifically be 2 nm, 4.6 nm, 6.8 nm or 7.8 nm. The material of the first spacing film **202** may be silicon boron carbon nitride, silicon nitride, or silicon carbonitride.

With reference to FIG. **28**, a third oxide film **203** is formed on the surface of the first spacing film **202**, and of the third oxide film **203**, a part higher than the top surface of the first spacing film **202** and a part at the bottom are removed through etching. A part of the third oxide film **203** on the side surface of the gate structure **210** is retained. In some embodiments, the third oxide film **203** is formed through the atomic layer deposition process. A thickness of the third oxide film **203** ranges from 2 to 10 nm, and may specifically be 2 nm, 3.8 nm, 6.2 nm, or 8.9 nm. The material of the third oxide film **203** may be silicon dioxide or silicon carbonitride.

With reference to FIG. **29**, a second spacing film **204** is formed on the surface of the third oxide film **203**, the first spacing film **202** and the gate structure **210**. A thickness of the second spacing film **204** ranges from 2 to 10 nm, and may specifically be 2 nm, 5.6 nm, 8.1 nm, or 9.8 nm. The material of the second spacing film **204** may be silicon boron carbon nitride, silicon nitride, or silicon carbonitride.

With reference to FIG. **30**, a fourth oxide film **205** is formed on the surface of the second spacing film **204**, and of the fourth oxide film **205**, a part higher than the top surface of the first spacing film **202** and a part at the bottom are removed through etching. A part of the fourth oxide film **205** on the side surface of the gate structure **210** is retained. In some embodiments, the fourth oxide film **205** is formed through the atomic layer deposition process. A thickness of the fourth oxide film **205** ranges from 2 to 10 nm, and may specifically be 2 nm, 3.8 nm, 6.2 nm, or 8.9 nm. The material of the fourth oxide film **205** may be silicon dioxide or silicon carbonitride.

With reference to FIG. **31**, a third spacing film **206** is formed on the surfaces of the second spacing film **204**, the fourth oxide film **205**, and the gate structure **210**, while a part of the second spacing film **204** and a part of the third spacing film **206** that are higher than the top of the first spacing film **202** are etched. A thickness of the third spacing film **206** ranges from 4 nm to 10 nm, and may specifically be 4 nm, 5.3 nm, 8.6 nm, or 9.9 nm. The material of the third spacing film **206** may be silicon boron carbon nitride, silicon nitride or silicon carbonitride.

In some embodiments, an etch selectivity between a material of the third oxide film **203** and a material of the first spacing film **202** is greater than 50. An etch selectivity between the material of the third oxide film **203** and a material of the second spacing film **204** is greater than 50. An etch selectivity between a material of the fourth oxide film **205** and the material of the first spacing film **202** is greater than 50. An etch selectivity between the material of the fourth oxide film **205** and the material of the second spacing film **204** is greater than 50.

With reference to FIG. **32**, the third oxide film **203** and the fourth oxide film **205** are removed through etching to form the cavity **240**. The retained first spacing film **202** is used as the first spacing layer **221**. The retained second spacing film **204** is used as the second spacing layer **222**. The retained third spacing film **206** is used as the third spacing layer **223**.

With reference to FIG. **33**, an isolation structure **250** is formed between adjacent two of the plurality of support structures. The isolation structure **250** includes a first isolation layer **251** and a second isolation layer **252** that are stacked. The materials of the first isolation layer **251** and the second isolation layer **252** may be one or more selected from the group consisting of silicon dioxide, silicon carbide, or silicon nitride.

Correspondingly, another embodiment of the present disclosure provides a method of manufacturing a semiconductor structure. A method of manufacturing a semiconductor structure provided by another embodiment of the present disclosure is substantially the same as the ones provided by the foregoing embodiments (FIGS. **4** to **25** or **26** to **33**). The main difference is that the given one of the plurality of first conductive structures provided by the foregoing embodiment of the present disclosure is a bit line, and the given one of the plurality of second conductive structures is a storage node contact structure, and the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures in the semiconductor structure provided by the another embodiment of the present disclosure are both metal wire structures. Details of contents or elements the same as or similar to those described in the foregoing embodiments are not repeated, and the ones different from the above are described in detail. The method of manufacturing a semiconductor structure provided by another embodiment of the present disclosure is described in detail below with reference to FIGS. **34** to **40**.

FIGS. **34** to **40** are schematic structural cross-sectional views corresponding to various steps of the method of manufacturing a semiconductor structure according to another embodiment of the present disclosure.

With reference to FIGS. **34** to **40**, the method of manufacturing a semiconductor structure includes: providing a base **300**; forming a plurality of first conductive structures and a plurality of second conductive structures, where the plurality of first conductive structures are located on a surface of the base **300** and distributed at intervals along a first direction X, and plurality of second conductive structures located on the surface of the base **300**, and the plurality of second conductive structures and the plurality of first conductive structures are arranged alternately; and forming a plurality of support structures located on the surface of the base **300** and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, where the given one of the plurality of support structures is provided with a support layer **320** therein arranged along the first direction X and a cavity **340** divided by the support layer **320**, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

With reference to FIG. **34**, a base **300** is provided and a plurality of metal wire structures **310** are formed on the surface of the base **300** and arranged at intervals along the first direction X.

In some embodiments, the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures are both metal wire structures **310**, and the metal wire structure **310** serve as the solder joints of the semiconductor structure. The support structures are located between adjacent metal wire structures **310**.

With reference to FIGS. **35** to **39**, the step of forming the plurality of support structures includes: sequentially forming a fourth spacing film **301**, a fifth oxide film **302**, a fifth spacing film **303**, and a sixth oxide film **304** on the base **300** and on top surfaces and side surfaces of the metal wire structures **310**; removing, through etching, a part of the fifth oxide film **302**, a part of the fifth spacing film **303**, and a part of the sixth oxide film **304** that are higher than a top of the fourth spacing film **301**; and removing the fifth oxide film **302** and the sixth oxide film **304**, to form the cavity, where the fourth spacing film **301** and a remaining part of the fifth spacing film **303** jointly constitute the support layer **320**.

In some embodiments, along the first direction X, the support layer **320** includes a fourth spacing layer **321** and a fifth spacing layer **322**. The fourth spacing layer **321** is located between the metal wire structure **310** and the cavity **340**. The fifth spacing layer **322** is located in the middle of the cavity **340**. The fourth spacing layer **321** is thinner than the fifth spacing layer **322**. The fifth spacing layer **322** in the cavity **340** is relatively thick, and can support the side surface of the cavity **340**, such that the support structure can withstand various stresses in the subsequent process, thereby avoiding deformation of the side surfaces of the cavity **340**, and enabling the structure of the cavity **340** to be complete. This may avoid that the K value increases to cause the parasitic capacitance between adjacent metal wire structures **310** to increase and lead to the short-circuit risk between them. A thickness of the fifth spacing layer **322** ranges from 4 nm to 10 nm, and may specifically be 4 nm, 5.3 nm, 8.6 nm, or 9.9 nm.

In some embodiments, along the first direction X, the thickness of the cavity **340** ranges from 2 nm to 20 nm. The cavity **340** includes a first cavity **341** and a second cavity **342** separated by the fourth spacing layer **322**. The thickness of the first cavity **341** and the thickness of the second cavity **342** both range from 2 nm to 10 nm, and may be specifically, 2 nm, 3.8 nm, 6.2 nm, or 8.9 nm.

With reference to FIG. **35**, a fourth spacing film **301** is formed on the surface of the base **300** and the top surface and side surface of the metal wire structure **310**. A thickness of the fourth spacing film **301** ranges from 2 to 8 nm, and may specifically be 2 nm, 4.6 nm, 6.8 nm, or 7.8 nm. The material of the fourth spacing film **301** may be silicon boron carbon nitride, silicon nitride, or silicon carbonitride.

With reference to FIG. **36**, a fifth oxide film **302** is formed on the surface of the fourth spacing film **301**, and of the fifth oxide film **302**, a part higher than the top surface of the fourth spacing film **301** and a part at the bottom are removed through etching. A part of the fifth oxide film **302** on the side surface of the metal wire structure **310** is retained. A thickness of the fifth oxide film **302** ranges from 2 to 8 nm, and may specifically be 2 nm, 4.6 nm, 6.8 nm, or 7.8 nm. The material of the fifth oxide film **302** may be silicon dioxide or silicon carbonitride.

With reference to FIG. **37**, a fifth spacing film **303** is formed on the fourth spacing film **301**, the fifth oxide film **302**, and the metal wire structure **310**. A thickness of the fifth spacing film **303** ranges from 4 to 10 nm, and may specifically be 4 nm, 5.3 nm, 8.6 nm, or 9.9 nm. The material of the fifth spacing film **303** may be silicon boron carbon nitride, silicon nitride, or silicon carbonitride.

With reference to FIG. **38**, a sixth oxide film **304** is formed on the surface of the fifth spacing film **303**, and of the sixth oxide film **304**, a part higher than the top surface of the fifth spacing film **303** and a part at the bottom are removed through etching. A part of the sixth oxide film **304** on the side surface of the metal wire structure **310** is retained. A thickness of the sixth oxide film **304** ranges from 2 to 8 nm, and may specifically be 2 nm, 4.6 nm, 6.8 nm, or 7.8 nm. The material of the sixth oxide film **304** may be silicon dioxide or silicon carbonitride.

With reference to FIG. **39**, the fifth oxide film **302** and the sixth oxide film **304** are removed through etching to form the cavity. The retained fourth spacing film **301** is used as the fourth spacing layer **321**. The retained fifth spacing film **303** is used as the fifth spacing layer **322**. A first isolation layer **351** is formed between adjacent support structures. Along the first direction X, a thickness of the first isolation layer **351** ranges from 5 nm to 30 nm. A material of the first isolation layer **351** may be one or more selected from the group consisting of silicon dioxide, silicon carbide or silicon nitride.

With reference to FIG. **40**, a second isolation layer **352** is formed on the top surfaces of the support structures, the top surfaces of the metal wire structures **310**, and the top surface of the first isolation layer **351**. The first isolation layer **351** and the second isolation layer **352** form the isolation structure **350**. A material of the second isolation layer **352** may be one or more selected from the group consisting of silicon dioxide, silicon carbide or silicon nitride.

Those of ordinary skill in the art can understand that the above implementations are specific embodiments for implementing the present disclosure; and in practical applications, various changes may be made in terms of forms and details without departing from the spirit and scope of the present disclosure. Any person skilled in the art may make changes and modifications without departing from the spirit and scope of the present disclosure. Therefore, the protection scope of the present disclosure should be subject to the scope defined by the claims.

The invention claimed is:

1. A semiconductor structure, comprising:
- a base;
- a plurality of first conductive structures, located on a surface of the base and distributed at intervals along a first direction;
- a plurality of second conductive structures, located on the surface of the base, and the plurality of second conductive structures and the plurality of first conductive structures being arranged alternately; and
- a plurality of support structures, located on the surface of the base and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, wherein the given one of the plurality of support structures is provided with a support layer therein arranged along the first direction and a cavity divided by the support layer, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

2. The semiconductor structure according to claim **1**, wherein the given one of the plurality of first conductive structures is a bit line, and the given one of the plurality of second conductive structures is a storage node contact structure; and the bit line extends along a second direction, the second direction intersects the first direction, the bit line comprises a conductive layer and a dielectric layer that are stacked, and a top surface of the support layer away from the base is not lower than a top surface of the conductive layer.

3. The semiconductor structure according to claim **2**, wherein along a direction perpendicular to the surface of the base, the support layer is as high as ⅓ to ⅔ of the bit line.

4. The semiconductor structure according to claim **2**, wherein along the first direction, the support layer comprises a first support layer, a second support layer, and a third support layer, the first support layer is located between the bit line and the cavity, the second support layer is located in a middle of the cavity, and the third support layer is located between the storage node contact structure and the cavity;

and a thickness of the first support layer or the third support layer is smaller than or equal to that of the second support layer.

5. The semiconductor structure according to claim **4**, wherein the thickness of the second support layer ranges from 2 nm to 4 nm.

6. The semiconductor structure according to claim **1**, further comprising: an isolation structure located between adjacent two of the plurality of support structures, wherein the given one of the plurality of support structures further extends along a side surface of the given one of the plurality of second conductive structures.

7. The semiconductor structure according to claim **6**, wherein the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures are both gate structures; and along the first direction, a thickness of the cavity ranges from 2 nm to 20 nm.

8. The semiconductor structure according to claim **7**, wherein along the first direction, the support layer comprises a first spacing layer, a second spacing layer, and a third spacing layer, the first spacing layer is located between the gate structure and the cavity, the second spacing layer is located in a middle of the cavity, and the third spacing layer is located between the isolation structure and the cavity; and a thickness of the first spacing layer or the third spacing layer is smaller than or equal to a thickness of the second spacing layer.

9. The semiconductor structure according to claim **6**, wherein the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures are both metal wire structures; and along the first direction, a thickness of the cavity ranges from 2 nm to 20 nm.

10. The semiconductor structure according to claim **9**, wherein along the first direction, a thickness of the isolation structure ranges from 5 nm to 30 nm.

11. The semiconductor structure according to claim **1**, wherein along the first direction, a ratio of a thickness of the cavity to a thickness of the support layer ranges from 1/3 to 4/3.

12. The semiconductor structure according to claim **1**, wherein a material of the support layer comprises silicon nitride, silicon carbonitride, or silicon boron carbon nitride.

13. A method of manufacturing a semiconductor structure, comprising:

providing a base;

forming a plurality of first conductive structures and a plurality of second conductive structures, wherein the plurality of first conductive structures are located on a surface of the base and are distributed at intervals along a first direction, and the plurality of second conductive structures are located on the surface of the base, and the plurality of second conductive structures and the plurality of first conductive structures are arranged alternately; and

forming a plurality of support structures located on the surface of the base and a given one of the plurality of support structures being located between a given one of the plurality of first conductive structures and a given one of the plurality of second conductive structures, wherein the given one of the plurality of support structures is provided with a support layer therein arranged along the first direction and a cavity divided by the support layer, and the given one of the plurality of support structures extends along a side surface of the given one of the plurality of first conductive structures.

14. The method of manufacturing a semiconductor structure according to claim **13**, wherein the given one of the plurality of first conductive structures is a bit line, and the given one of the plurality of second conductive structures is a storage node contact structure; and the forming a plurality of support structures comprises:

sequentially forming a first support film, a first oxide film, a second support film, a second oxide film, and a third support film on the base and on top surfaces and side surfaces of the plurality of first conductive structures;

removing, through etching, the first support film, the first oxide film, the second support film, the second oxide film, and the third support film by a thickness, to expose the plurality of first conductive structures with a thickness; and

removing, through etching, the first oxide film and the second oxide film, to form the cavity, wherein remaining parts of the first support film, second support film, and third support film form the support layer.

15. The method of manufacturing a semiconductor structure according to claim **14**, wherein an etch selectivity between a material of the first oxide film and a material of the first support film is greater than 50; and an etch selectivity between the material of the first oxide film and a material of the second support film is greater than 50.

16. The method of manufacturing a semiconductor structure according to claim **14**, wherein an etch selectivity between a material of the second oxide film and the material of the first support film is greater than 50; and an etch selectivity between the material of the second oxide film and the material of the second support film is greater than 50.

17. The method of manufacturing a semiconductor structure according to claim **13**, wherein the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures are both gate structures; and the forming a plurality of support structures comprises:

sequentially forming a first spacing film, a third oxide film, a second spacing film, a fourth oxide film, and a third spacing film on the surface of the base as well as top surfaces and side surfaces of the gate structures;

removing, through etching, a part of the third oxide film, a part of the second spacing film, a part of the fourth oxide film, and a part of the third spacing film that are higher than a top of the first spacing film; and

removing the third oxide film and the fourth oxide film through etching to form the cavity, wherein remaining parts of the first spacing film, second spacing film, and third spacing film form the support layer.

18. The method of manufacturing a semiconductor structure according to claim **13**, wherein the given one of the plurality of first conductive structures and the given one of the plurality of second conductive structures are both metal wire structures; and the forming a plurality of support structures comprises:

sequentially forming a fourth spacing film, a fifth oxide film, a fifth spacing film, and a sixth oxide film on the base and on top surfaces and side surfaces of the metal wire structures;

removing, through etching, a part of the fifth oxide film, a part of the fifth spacing film, and a part of the sixth oxide film that are higher than a top of the fourth spacing film; and

removing the fifth oxide film and the sixth oxide film, to form the cavity, wherein the fourth spacing film and a remaining part of the fifth spacing film jointly constitute the support layer.

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